MODEL QUESTION PAPER

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THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

CHEMICAL PROCESS PRINCIPLES

Time-3hrs Max.Marks:100

PART-A

I Answer all questions in one or two sentences. Each question carries 2 marks

1) What is molarity?
2) Write the composition of atmospheric air?
3) Write two examples of tie element
4) Define selectivity
5) Define entropy 2x5= 10

PART-B

II (Answer any five questions. Each question carries 6 marks)

1) Explain the various system of units 6
2) A Sulphuric acid solution has a molarity of 11.24 and molality 94 Calculate the density of solution 6
3) Draw the block diagram and write the material balance equations for distillation and evaporation? 6
4) Soyabeen seeds are extracted with hexane in batch extractors . the flaked seed contain 18.6% oil , 69% solids, and 12.4% moisture. At the end of the extraction process cake is separated from the hexane-oil mixture. The analysis yields 0.8% oil, 87.7% solids and11.8% moisture. Find the % recovery of oil? 6
5) The Carbon monoxide (CO) reacted with Hydrogen to produce methanol .Calculate from the reaction (a) The stoichiometric ratio of H_2 to CO (b) Kmol of CH_3OH produced per Kmol of CO reacted (c) the weight ratio of CO to H_2 both are fed to reactor in stoichiometric proportion. 6
6) A coal has ultimate composition C:67.34%,H2:4.67%,O_2:8.47% N2:1.25% and S :4.77% and the rest is ash. Find the theoretical air-fuel ratio 6
7) State Hess's law of heat of summation with an example 6

PART-C

(Answer one full question from each module. Each question carries 15 marks)

MODULE-I

III 1 In the manufacture of nitric acid initially ammonia and air are mixed at 7 atm and 650^0C the composition of the mixture in volume basis is as follows.

Nitrogen - 70.5%

Oxygen - 18.8% 8

Water - 1.2%
Ammonia - 9.5%
Find a) the% composition by mass ? b)density of gas mixture using ideal gas law.

2 5 kg of Oxygen contained in closed container of volume $1m^3$ is heated with out exceeding a pressure of 709.21kpa.Calculate maximum temperature of gas attained. 7

OR

IV 1 The analysis of magnesite ore yields 81%MgCO3.14% SiO2 and 5% H2O(by weight) Convert the analysis into mole%. 7

2 A sample of wine contain 20% alcohol(ethanol) on volume basis .Find the weight % of alcohol in the wine .Assume the densities of alcohol and alcohol free liquid (essentially water) to be $0.79 gm/cm^3$ and $1 gm/cm^3$ 8

MODULE-II

V 1 A 100 kg mixture consisting of 27.8% Acetone (A) and 72.2% of Chloroform (B) by weight is to be batch extracted with a mixed solvent at 25^0C.the mixed solvent of unknown composition is known to contain water (S_1) and Acetic acid (S_2) .the mixture of the original mixture and mixed solvent is shaken well, allowed to attain equilibrium and separated into two layers. The composition of the two layers are given below 8

layer	Composition wt%			
	A	B	S_1	S_2
Upper layer(x)	7.5	3.5	57.4	31.6
Lower layer(y)	20.3	67.3	2.8	9.6

Find (a) the quantities of the two layers (b) the weight ratio of the mixed solvent to the original mixture

2. Soyabeen seeds are extracted with hexane in batch extractors . the flaked seed contain 18.6% oil , 69% solids, and 12.4% moisture. At the end of the extraction process cake is separated from the hexane-oil mixture. The analysis yields 0.8% oil, 87.7% solids and11.8% moisture. Find the % recovery of oil? 7

OR

VI 1 The analysis of sample of Babul bark yields 5.8% moisture ,12.6% tannin 8.3% soluble non tannin organic matter and the rest lignin . In order to extract tannin out of the bark ,a counter current extraction process is employed. The residue from the extraction process is analyzed and found to contain 0.92% tannin and 0.65% soluble non tannin organic matter on a dry basis. Find the percentage of tannin recovered on the basis of the original tannin present in the bark. All analyses are given on a weight basis. 8

2 Draw the block diagram and the material balance equations for crystallization and evaporation 7

MODULE-III

VII 1 Mono chloro acetic acid(MCA) is manufactured in a semi batch reactor by the action of glacial acetic acid with chlorine gas at 373k(100^0C) in the presence of Phosphorous tri chloride(PCl_3)as a catalyst. MCA thus formed will further react with chlorine to form Di chloro acetic acid (DCA). To prevent the formation of DCA, excess acetic acid used. A small scale unit which produces 5000 Kg/day MCA requires 4536 kg/day of chlorine gas. Also 263 kg/day of DCA is separated from the crystallizer to get almost pure MCA product .Find the % of conversion and % of yield of MCA. 8

2 The Carbon monoxide (CO) reacted with Hydrogen to produce methanol .Calculate from the reaction (a) The stoichiometric ratio of H_2 to CO (b) Kmol of CH_3OH produced per Kmol of CO reacted 7

OR

VIII 1 A combustion reaction is fed with 50kg mol/hr of butane and2000kg mol/hr of air. Calculate the % excess air used and composition of gases Leaving the combustion reactor. Assuming complete combustion of butane 8

C_4H_{10}+ $13/2O_2 \rightarrow 4CO_2+5H_2O$

2 Explain the recycling operation with a diagram 7

MODULE-IV

IX

1 Given the heat of formation of CO,CO_2 and H_2O (g) are -110.5,-393.8 and -241.8 kJ/mol respectively, calculate the heat of reaction: 7

$CO_2+H_2 \rightarrow CO + H_2O$

Given data :

(1) $C(s) + 0.5\ O_2 \rightarrow CO\ (g)$ ΔH = -110.5kJ

(2) $C(g) + O_2 \rightarrow CO2(g)$ ΔH =--393.8kJ

(3) $H_2(g) + 0.5O_2(g) = H_2O(l)$ ΔH = -241.8kJ

2 Explain the term

(i)Enthalpy and derive $\Delta H = \Delta E$ 8

(ii) first law of thermodynamics and its mathematical formulation

OR

X 1 The heat of combustion of ethyl alcohol is-33000 calories. If the heat of formation of carbon dioxide and water are -94300and -68500 calories respectively .Calculate the heat of formation of ethyl alcohol.

Given data: 8

(i) $C_2H_5OH\ (l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$ $\Delta H = -330000$ calories

(ii) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -943000$ calories

(iii) $H_2(g) + 0.5O_2 \rightarrow H_2O(l)$ $\Delta H = -68500$ calories

2 State about heat capacity and derive the relation between $Cp = Cv$ 7
